

Transformers in Practice: BERT, GPT, and Beyond

Scaler School of Technology | Deep Learning I | Week 7 — Session 2/2 | Post-Lecture Review & Practice



Key Takeaways

1 Pre-training + Fine-tuning

Pre-train a large model on massive unlabeled text (learn language patterns), then fine-tune on a small labeled dataset for your specific task. This paradigm made training task-specific models from scratch obsolete.

2 BERT vs GPT — Two Sides of the Same Coin

BERT uses bidirectional attention (encoder-only, MLM training) and excels at understanding tasks (classification, NER, QA). GPT uses causal attention (decoder-only, next-token prediction) and excels at generation. The **only** architectural difference is the attention mask.

3 Tokenization Determines Model Capabilities

Subword tokenization (BPE/WordPiece) splits rare words into frequent pieces. Numbers get tokenized inconsistently (“427” → “42” + “7”), which is why LLMs hallucinate on arithmetic. The tokenizer is the invisible bottleneck of every NLP system.

Test Your Understanding

Try answering these without looking at the slides. Answers are on page 3.

- Q1.** Why can't BERT generate text, even though it “understands” language so well?
- Q2.** You have 500 labeled customer support emails. Should you train a text classifier from scratch or fine-tune BERT? Why?
- Q3.** Explain why doubling the input sequence length quadruples the memory cost of self-attention.
- Q4.** GPT-3 can translate languages with just 2–3 examples in the prompt and *no* weight updates. What is this ability called, and why is it surprising?
- Q5.** A tokenizer splits “15213” into [“152”, “13”]. How does this cause arithmetic errors in LLMs?

Work through these in a Colab notebook. No starter code — figure out the approach from what you learned.

Exercise 1: BERT Masked Token Prediction

Load `bert-base-uncased` via HuggingFace. Create 5 sentences with [MASK] at ambiguous words (e.g., “The bank was [MASK]”). Print top-5 predictions per mask. **Output:** table of input sentence, masked position, top-5 tokens with probabilities.

Exercise 2: GPT-2 Temperature Experiment

Load `gpt2`. Generate 50-token completions for “The future of artificial intelligence is” at temperatures 0.2, 0.7, 1.5 (3 runs each). **Output:** 9 texts + 1-paragraph analysis of how temperature affects coherence vs. creativity.

Exercise 3: Tokenizer Comparison

Load GPT-2 and BERT tokenizers. Tokenize: “unhappiness”, “counterintuitively”, “12345”, “pneumonoultramicroscopicsilicovolcanoconiosis”. **Output:** comparison table of token counts and subword splits per tokenizer.

Challenge Problem

Fine-tune BERT for Sentiment Classification (end-to-end)

Build a complete sentiment classifier: (1) Load IMDB dataset (2000-sample subset via `datasets` library), (2) Tokenize with `BertTokenizer` (`max_length=128`, padding/truncation), (3) Load `BertForSequenceClassification` (`num_labels=2`), (4) Train 3 epochs with `lr=2e-5`, AdamW, (5) Report test accuracy.

Bonus: Compare randomly-initialized BERT vs. pre-trained BERT on the same 2000 samples. How much does pre-training help?

- **BERT:** Devlin et al. (2019), “*BERT: Pre-training of Deep Bidirectional Transformers.*” — Focus on Sec. 3 (pre-training) and Sec. 4 (fine-tuning).
- **GPT-2:** Radford et al. (2019), “*Language Models are Unsupervised Multitask Learners.*” — Zero-shot performance; read intro + results.
- **GPT-3:** Brown et al. (2020), “*Language Models are Few-Shot Learners.*” — Scale unlocks emergent abilities; focus on Sec. 3.
- **BPE:** Sennrich et al. (2016), “*Neural Machine Translation of Rare Words with Subword Units.*” — Short, clear, worth reading in full.

Online Resources & Tools

- **HuggingFace Hub** — Pre-trained models, tokenizers, and datasets ready to use
- **Jay Alammar’s Blog** — Visual guides to BERT, GPT, and Transformers
- **OpenAI Tokenizer** — Interactive BPE tokenizer playground (platform.openai.com/tokenizer)
- **BERT Visualizer** — exBERT interactive attention explorer
- **Papers With Code** — Leaderboards for NLP benchmarks (GLUE, SuperGLUE)
- **Andrej Karpathy’s nanoGPT** — Minimal GPT implementation for learning

Fill the Gaps — Prerequisite Refreshers

Felt lost on a specific concept? These resources will help you catch up.

- **Confused about attention masks?** Review Week 7 Session 1 slides — full attention (bidirectional) vs. causal masking (left-to-right only).
- **Unclear on softmax temperature?** $\text{softmax}(z_i/T)$: low T = peaked/deterministic, high T = flat/random. Try with 3 numbers in a calculator.
- **What’s fine-tuning vs. training from scratch?** Fine-tuning starts from pre-trained weights (small l_r , few epochs). From scratch = random init, needs $100\times$ more data.
- **Forgot cross-entropy loss?** Measures distance between predicted probabilities and true label. Used in both MLM and classification. Review Week 3.

Answers — Test Your Understanding

- A1.** BERT sees all positions simultaneously (bidirectional). To generate token 5, it would peek at tokens 6+. No causal ordering means no sequential generation.
- A2.** Fine-tune BERT. 500 labels is far too few to train from scratch (would overfit). BERT already knows language — just teach it your task. Expect 85%+ accuracy.
- A3.** $QK^\top$ is $n \times n$. Double $n \rightarrow 2n \times 2n = 4n^2$ entries. Quadratic scaling.
- A4.** *Few-shot learning* (in-context learning). Surprising because there are no weight updates — pattern recognition purely from prompt examples.
- A5.** The model sees “152” and “13” as separate units, not digits 1-5-2-1-3. It operates on inconsistent chunks, so arithmetic on the underlying digits fails.

Hands-on notebook: `bert_gpt_notebook.ipynb`

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